Classical versus Deep Learning for Paediatric Pneumonia Screening

Spoken script for the Machine Learning (PCOM7E) individual presentation, one section per slide.

Slide 1: Title

Hello. In this talk I compare two ways of deciding from a child's chest X-ray whether it shows pneumonia: a classical pipeline, where I choose the image features and a support vector machine learns to weigh them, and a convolutional neural network, which sees only pixels and discovers its own features. I will cover the data, validation and results, the evidence each model relies on, who it fails and how it could be deployed, ending with a demonstration.

Slide 2: Why paediatric pneumonia

The stakes here are real and uneven. World Health Organization figures show that pneumonia killed roughly 740,000 children under five in 2019 and that just one child in three with pneumonia is given the antibiotics it requires (World Health Organization, 2022). Bacterial cases need urgent antibiotics, while viral cases receive supportive care (Kermany et al., 2018), so a missed pneumonia and a false alarm do not cost the same.

Slide 3: The dataset and where it came from

The radiographs are of children aged one to five at a hospital in Guangzhou (Kermany et al., 2018). I chose the original Mendeley Data release of 5,856 images because it is physician-graded, openly licensed under CC BY 4.0 and keeps the authors' fixed split of 5,232 training and 624 test images (Kermany, Zhang and Goldbaum, 2018). Some children appear up to thirty times, but identity is recorded only in the file name, so I treat that number as the patient.

Slide 4: Preprocessing

Every image is converted to greyscale, resized to 128 by 128 pixels and scaled between zero and one, which keeps the lung fields and keeps training feasible. The classical model never sees pixels: each image becomes a histogram of oriented gradients, 1,764 numbers describing edge direction and strength, a descriptor that Dalal and Triggs (2005) showed to outperform earlier features for detecting people. Both models weight the classes equally, since pneumonia outnumbers normal images three to one.

Slide 5: Model design

On the classical side I used the scikit-learn toolkit, whose pipelines refit the scaling inside each training fold (Pedregosa et al., 2011), to train an RBF-kernel support vector machine, its penalty tuned by cross-validation. The deep model is a compact four-layer network in TensorFlow and Keras, chosen for its augmentation layers and the gradient access Grad-CAM needs, and trained from scratch so the comparison is fair. Its filters double as the image shrinks, with Adam at the standard rate of 0.001, batches of 32, 30 per cent dropout, at most 20 epochs and no flips, as mirroring

misplaces the heart. In development, batch normalisation and over-eager early stopping each broke two folds, so I switched to group normalisation and delayed early stopping.

Slide 6: Validation strategy

Validation estimates performance on children a model has never seen. A single split gives one estimate that repeated tuning can quietly overfit, whereas five folds give five estimates and their spread. Shuffling images into folds would put around 480 children on both sides of every split, so I grouped the folds by patient while keeping the class balance. Kohavi (1995) judged stratified folds the most dependable basis for model selection on real-world data and favoured ten, although I used five for practicality. The test set was never used for tuning.

Slide 7: Cross-validation results

In cross-validation the classical model leads, with 97.7 per cent accuracy against 85.5 and an F1 of 0.98 against 0.88. The network’s spread is revealing: it ranks cases well in every fold, with a mean ROC area of 0.97, but in two folds its scores sit so low that at 0.5 it misses many pneumonias. Its training and monitoring losses fall together in three folds, but in two the monitoring loss levels off well above, suggesting partial memorising. The 97.7 per cent should still prompt suspicion, since a result this high on medical images can reflect a shortcut rather than skill.

Slide 8: The test set reverses the ranking

On the 624 radiographs the authors held back, the order flips. At the default threshold the network reaches 86.1 per cent accuracy against 77.7 and an F1 of 0.90 against 0.85. Recall holds up, but specificity falls from 96 and 97 per cent to 41 for the support vector machine and 69 for the network: the test normals differ from the training normals, and both models read that difference as disease. Kermany and colleagues kept 90 per cent specificity on these images with a pretrained network (Kermany et al., 2018), so transfer learning is the next comparison. Independence is uncertain: 170 pneumonia identifiers appear in both folders.

Slide 9: Choosing an operating point

Because the errors differ in cost, I also chose a recall-led threshold on the out-of-fold predictions, keeping recall at 95 per cent. For the support vector machine, 0.91 lifted test specificity to 64 per cent, but the network’s two poorly calibrated folds dragged its cut to 0.11, leaving 32 per cent specificity. Hand-set thresholds can become invalid once a model or its data change (Sculley et al., 2015), as happened here. On precision against recall, which Saito and Rehmsmeier (2015) favour when classes are unbalanced, the network stays ahead until recall passes 95 per cent.

Slide 10: What the network attends to

I used Grad-CAM, which weights the last convolutional layer’s feature maps by how strongly each moves the prediction and turns the result into a heat map (Selvaraju et al., 2017). About a third of the attention falls in the outer border, against 44 per cent if

spread evenly. Averaged over pneumonia films the attention sits over the heart and mediastinum, and over normal films on the shoulders and diaphragm, rather than the lungs. Central is not the same as clinically meaningful.

Slide 11: Comparing both models with LIME

Grad-CAM only works for the network, so to compare both models I used LIME, which switches image regions off, watches the prediction change and fits a simple local explanation (Ribeiro, Singh and Guestrin, 2016). It is model-agnostic and explains through regions a radiologist can recognise, about fifty-six per image. Across forty test images the support vector machine draws on the lung fields, but the network's importance gathers below the diaphragm, contradicting Grad-CAM. An explanation is therefore evidence to question, not proof of what a model has learned.

Slide 12: Subgroups

The dataset records no sex, ethnicity or social background, so I derived two attributes. By subtype there is no meaningful gap: the network detects 97 per cent of bacterial and 95 per cent of viral cases, which matters because bacterial cases are urgent. By exposure, the nearest equivalent to lighting conditions, recall stays high, but the support vector machine clears fewer normal images as they brighten, from 47 to 38 per cent. The intervals overlap, so this is a pattern to monitor, not a proven bias.

Slide 13: An acquisition shortcut

Since exposure explained little, I examined how images were stored. Normal radiographs were saved at about 1.7 times the height of pneumonia ones. A classifier given only height, width and shape, with no pixels, scored 83 per cent in cross-

validation, against 74 for always answering pneumonia, and 83 on the test set, above the support vector machine’s 78 and close to the network’s 86. As normal films widen, the support vector machine’s false alarms rise from a third to about 80 per cent, even after cropping to a square, and the network’s from 15 to 58. Acquisition differences, perhaps equipment, survive cropping; global edge histograms likely capture them. Brown et al. (2023) tie such shortcut learning to unequal performance across patient groups, and Zech et al. (2018) point to the safeguard: validating on hospitals the model never trained on.

Slide 14: Ethical considerations

Three ethical issues arise. First, privacy and provenance: the images come from routine care with ethics approval and privacy-compliant handling, but consent from families is not described (Kermany et al., 2018), and a reuse licence is not consent. The files carry only a numeric patient code, yet some films show burned-in text, so de-identification would need checking image by image. Second, unequal errors: a false negative can delay antibiotics for a sick child; a false positive costs a radiologist’s time. At the default threshold the network misses 15 pneumonia radiographs and flags 72 normal ones. Third, bias: chest X-ray classifiers have consistently underdiagnosed under-served groups (Seyyed-Kalantari et al., 2021), which one hospital and one age band cannot audit.

Slide 15: Recommendation and deployment

I recommend the network, because it generalised better, leaned less on the shortcut and gives each prediction a heat map a radiologist can challenge. The support vector machine is cheaper, cross-validating in two minutes rather than half an hour and running about seven times faster, but at fifteen milliseconds per image both are

negligible. I would deploy the network only as a triage aid: images de-identified on site, scored behind an inference API on a cloud platform such as Vertex AI or SageMaker and returned to the worklist with the heat map, the radiologist keeping the final read. The endpoint would scale out with volume, and every retrained model or new threshold would be re-validated before release. Monitoring (Sculley et al., 2015) would track recall and specificity by subgroup and watch the geometry signals.

Slide 16: Demonstration

The notebook loads both saved models and screens test radiographs neither was trained on. The normal case scores low on both, and both pneumonias are flagged with near certainty. The fourth, a bacterial pneumonia, is flagged with certainty by the support vector machine but scores 0.25 on the network and would be missed. At the network's recall-led threshold of 0.11 it is flagged, which is my argument: where the threshold sits is a clinical judgement rather than a technical setting.

Slide 17: Conclusion

In conclusion, cross-validation favoured the classical model, but the held-out test set favoured the network, which generalised better and leaned less on a shortcut, although its scores shifted between folds. Grad-CAM and LIME disagreed, so neither can be trusted alone. Above all, a model using no pixels outscored the support vector machine on test, exposing a confound in how the data were collected. The network is the better candidate, but only as a monitored triage aid, validated on unseen hospitals before it is trusted with a single child. Thank you for watching.

Slide 18: References

The full reference list is shown on this slide.

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